JEE Mains 2019 Chapter wise Question Bank

Area Under Curves – Questions

Q1

The area (in sq. units) bounded by the parabola $y = x^2 - 1$, the tangent at the point (2, 3) to it and the y-axis is:

(1) $\frac{8}{3}$ (2) $\frac{32}{3}$

(3) $\frac{56}{3}$ (4) $\frac{14}{3}$

9 Jan Morning

Q2

The area of the region
$A = \{(x, y): 0 \le y \le x\,|x| + 1 \text{ and } -1 \le x \le 1\}$ in sq. units is:

(1) $\frac{2}{3}$ (2) 2

(3) $\frac{4}{3}$ (4) $\frac{1}{3}$

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Q3

If the area enclosed between the curves $y = kx^2$ and $x = ky^2$, $(k > 0)$, is 1 square unit. Then k is:

(1) $\frac{\sqrt{3}}{2}$ (2) $\frac{1}{\sqrt{3}}$

(3) $\sqrt{3}$ (4) $\frac{2}{\sqrt{3}}$

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Q4

The area (in sq. units) of the region bounded by the curve $x^2 = 4y$ and the straight line $x = 4y - 2$ is :

(1) $\frac{5}{4}$ (2) $\frac{9}{8}$

(3) $\frac{7}{8}$ (4) $\frac{3}{4}$

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Q5

The area (in sq. units) in the first quadrant bounded by the parabola, $y = x^2 + 1$, the tangent to it at the point (2, 5) and the coordinate axes is :

(1) $\frac{8}{3}$ (2) $\frac{37}{24}$

(3) $\frac{187}{24}$ (4) $\frac{14}{3}$

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Q6

The area (in sq. units) of the region bounded by the parabola, $y = x^2 + 2$ and the lines, $y = x + 1$, $x = 0$ and $x = 3$, is :

(1) $\frac{15}{4}$ (2) $\frac{21}{2}$

(3) $\frac{17}{4}$ (4) $\frac{15}{2}$

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Q7

The area (in sq. units) of the region
$A = \{(x, y) \in R \times R \mid 0 \le x \le 3, 0 \le y \le 4, y \le x^2 + 3x\}$ is :

(1) $\frac{53}{6}$ (2) 8 (3) $\frac{59}{6}$ (4) $\frac{26}{3}$

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Q8

Let $S(\alpha) = \{(x, y) : y^2 \leq x, 0 \leq x \leq \alpha\}$ and $A(\alpha)$ is area of the region $S(\alpha)$. If for a λ, $0 < \lambda < 4$, $A(\lambda) : A(4) = 2 : 5$, then λ equals :

(1) $2\left(\frac{4}{25}\right)^{\frac{1}{3}}$ (2) $2\left(\frac{2}{5}\right)^{\frac{1}{3}}$

(3) $4\left(\frac{2}{5}\right)^{\frac{1}{3}}$ (4) $4\left(\frac{4}{25}\right)^{\frac{1}{3}}$

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Q9

The area (in sq. units) of the region $A = \{(x, y) : x^2 \leq y \leq x + 2\}$ is:

(1) $\frac{10}{3}$ (2) $\frac{9}{2}$ (3) $\frac{31}{6}$ (4) $\frac{13}{6}$

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Q10

The area (in sq. units) of the region

$$A = \left\{(x, y) : \frac{y^2}{2} \leq x \leq y + 4\right\} \text{ is:}$$

(1) $\frac{53}{3}$ (2) 30 (3) 16 (4) 18

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Q11

The region represented by $|x - y| \leq 2$ and $|x + y| \leq 2$ is bounded by a :

(1) square of side length $2\sqrt{2}$ units

(2) rhombus of side length 2 units

(c) square of area 16 sq. units

(4) rhombus of area $8\sqrt{2}$ sq. units

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Q12

The area (in sq. units) of the region bounded by the curves $y = 2^x$ and $y = |x + 1|$, in the first quadrant is :

(1) $\log_e 2 + \frac{3}{2}$ (2) $\frac{3}{2}$

(3) $\frac{1}{2}$ (4) $\frac{3}{2} - \frac{1}{\log_e 2}$

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Q13

If the area (in sq. units) of the region $\{(x, y) : y^2 \leq 4x, x + y \leq 1, x \geq 0, y \geq 0\}$ is $a\sqrt{2} + b$, then $a - b$ is equal to :

(1) $\frac{10}{3}$ (2) 6 (3) $\frac{8}{3}$ (4) $-\frac{2}{3}$

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Q14

If the area (in sq. units) bounded by the parabola $y^2 = 4\lambda x$ and the line $y = \lambda x$, $\lambda > 0$, is $\frac{1}{9}$, then λ is equal to :

(1) $2\sqrt{6}$ (2) 48 (3) 24 (4) $4\sqrt{3}$

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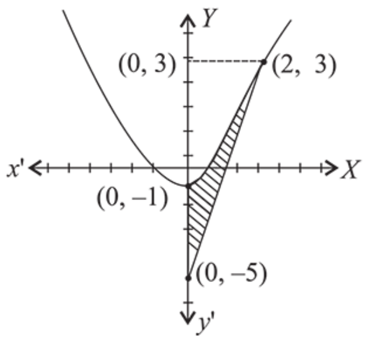
JEE Mains 2019 Chapter wise Question Bank

Area Under Curves - Solutions

Q1

(1)



$\because$ Curve is given as :
$y = x^2 - 1$

$\Rightarrow \quad \dfrac{dy}{dx} = 2x$

$\Rightarrow \quad \left(\dfrac{dy}{dx}\right)_{(2,3)} = 4$

equation of tangent at (2, 3)

$(y - 3) = 4(x - 2)$

$\Rightarrow y = 4x - 5$

but $x = 0$

$\Rightarrow y = -5$

Here the curve cuts Y–axis

$\therefore$ required area

$$= \frac{1}{4}\int_{-5}^{3}(y+5)dy - \int_{-1}^{3}\sqrt{y+1}\,dy$$

$$= \frac{1}{4}\left[\frac{y^2}{2} + 5y\right]_{-5}^{3} \frac{-2}{3}\left[(y+1)^{3/2}\right]_{-1}^{3}$$

$$= \frac{1}{4}\left[\frac{9}{2} + 15 - \frac{25}{2} + 25\right] - \frac{2}{3}\left[4^{3/2} - 0\right]$$

$$= \frac{32}{4} - \frac{16}{3} = \frac{8}{3} \text{ sq-units.}$$

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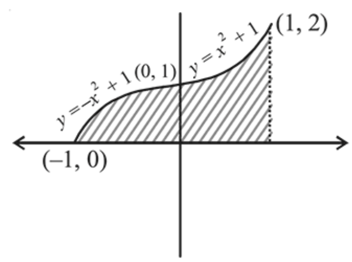
Q2

(2) Given $A = \{(x, y) : 0 \le y \le x|x| + 1 \text{ and } -1 \le x \le 1\}$



$\therefore$ Area of shaded region

$$= \int_{-1}^{0}(-x^2+1)dx + \int_{0}^{1}(x^2+1)dx$$

$$= \left(-\frac{x^3}{3} + x\right)_{-1}^{0} + \left(\frac{x^3}{3} + x\right)_{0}^{1}$$

$$= 0 - \left(\frac{1}{3} - 1\right) + \left(\frac{1}{3} + 1\right) - (0 + 0)$$

$$= \frac{2}{3} + \frac{4}{3} = \frac{6}{3} = 2 \quad \text{square units}$$

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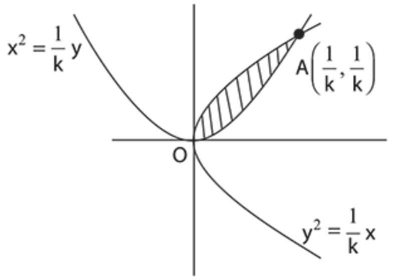
Q3

(2)



Two curves will intersect in the Ist quadrant at $A\left(\frac{1}{k}, \frac{1}{k}\right)$

$\because$ area of shaded region = 1.

$$\therefore \int_0^{\frac{1}{k}} \left(\frac{\sqrt{x}}{\sqrt{k}} - kx^2\right) dx = 1$$

$$\Rightarrow \left(\frac{1}{\sqrt{k}} \cdot \frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right)_0^{\frac{1}{k}} - \left(k \cdot \frac{x^3}{3}\right)_0^{\frac{1}{k}} = 1$$

$$\Rightarrow \frac{2}{3\sqrt{k}} \cdot \frac{1}{k^{\frac{3}{2}}} - \frac{k}{3k^3} = 1$$

$$\Rightarrow \frac{2}{3k^2} - \frac{1}{3k^2} = 1$$

$$\Rightarrow 3k^2 = 1$$

$$\Rightarrow k = \pm \frac{1}{\sqrt{3}}$$

$$\therefore k = \frac{1}{\sqrt{3}} \qquad (\because k > 0)$$

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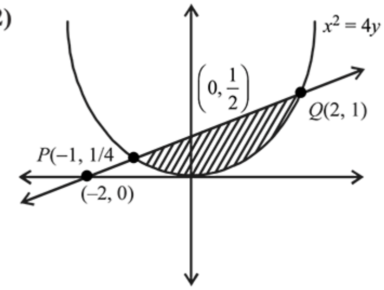
Q4

(2)



Let points of intersection of the curve and the line be P and Q

$$x^2 = 4\left(\frac{x+2}{4}\right)$$

$$x^2 - x - 2 = 0$$

$$x = 2, -1$$

Point are (2, 1) and $\left(-1, \frac{1}{4}\right)$

$$\text{Area} = \int_{-1}^{2} \left[\left(\frac{x+2}{4}\right) - \left(\frac{x^2}{4}\right)\right] dx = \left[\frac{x^2}{8} + \frac{1}{2}x - \frac{x^3}{12}\right]_{-1}^{2}$$

$$= \left(\frac{1}{2} + 1 - \frac{2}{3}\right) - \left(\frac{1}{8} - \frac{1}{2} + \frac{1}{12}\right) = \frac{9}{8}$$

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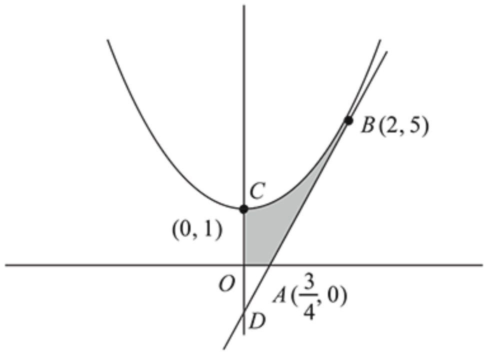
Q5

(2)



The equation of parabola $x^2 = y - 1$

The equation of tangent at (2, 5) to parabola is

$$y - 5 = \left(\frac{dy}{dx}\right)_{(2,5)} (x - 2)$$

$$y - 5 = 4(x - 2)$$

$$4x - y = 3$$

Then, the required area

$$= \int_0^2 \{(x^2+1) - (4x-3)\}\ dx - \text{Area of } \Delta AOD$$

$$= \int_0^2 (x^2 - 4x + 4)\ dx - \frac{1}{2} \times \frac{3}{4} \times 3$$

$$= \left[\frac{(x-2)^3}{3}\right]_0^2 - \frac{9}{8} = \frac{37}{24}$$

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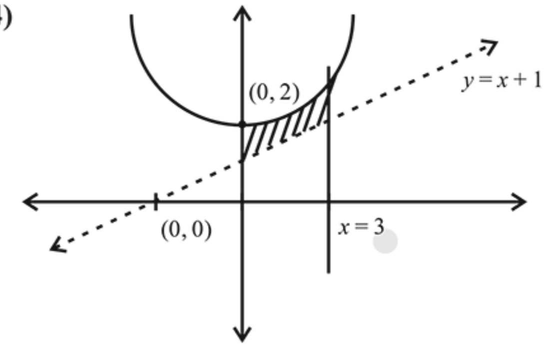
Q6

(4)



Area of the bounded region $\int_0^3 [(x^2+2) - (x+1)]\,dx$

$$= \left[\frac{x^3}{3} - \frac{x^2}{2} + x\right]_0^3$$

$$= 9 - \frac{9}{2} + 3 = \frac{15}{2}$$

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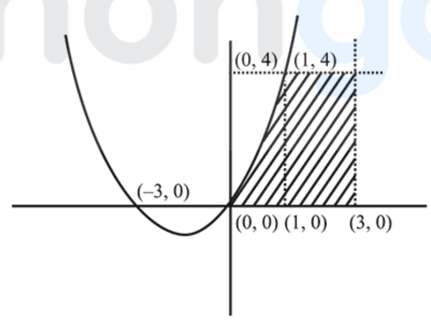
Q7

(3) Since, the relation $y \le x^2 + 3x$ represents the region below the parabola in the 1st quadrant



$\because\ y = 4$

$\Rightarrow\ x^2 + 3x = 4 \Rightarrow x = 1, -4$

$\therefore$ the required area = area of shaded region

$$= \int_0^1 (x^2 + 3x)dx + \int_1^3 4.dx \quad = \left[\frac{x^3}{3} + \frac{3x^2}{2}\right]_0^1 + [4x]_1^3$$

$$= \frac{1}{3} + \frac{3}{2} + 8 = \frac{59}{6}$$

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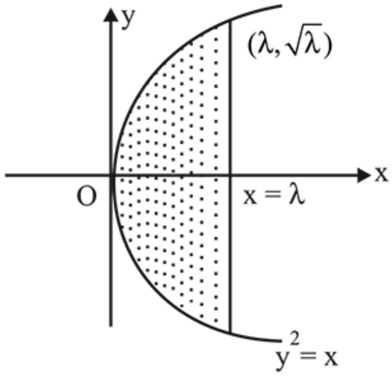
Q8

(4)



Area of the region = $2\times\int_0^\lambda y\,dx = 2\int_0^\lambda \sqrt{x}\,dx$

$= 2\times\frac{2}{3}\pi^{\frac{3}{2}}$

$A(\lambda) = 2\times\frac{2}{3}(\lambda\times\sqrt{\lambda}) = \frac{4}{3}\lambda^{\frac{3}{2}}$

Given, $\frac{A(\lambda)}{A(4)} = \frac{2}{5} \Rightarrow \frac{\lambda^{\frac{3}{2}}}{8} = \frac{2}{5}$

$\lambda = \left(\frac{16}{5}\right)^{\frac{2}{3}} = 4.\left(\frac{4}{25}\right)^{\frac{1}{3}}$

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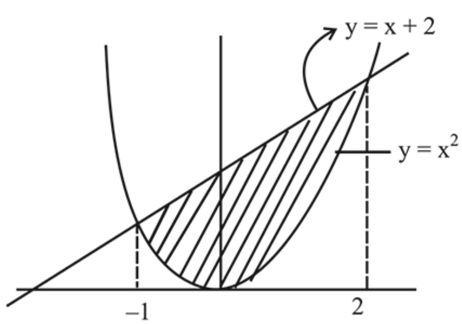
Q9

(2)



Required area is equal to the area under the curves $y \geq x^2$ and $y \leq x+2$

$\therefore$ requried area $\int_{-1}^{2}((x+2)-x^2)dx$

$= \left(\frac{x^2}{2}+2x-\frac{x^3}{3}\right)_{-1}^{2}$

$= \left(2+4-\frac{8}{3}\right)-\left(+\frac{1}{2}-2+\frac{1}{3}\right) = \frac{9}{2}$

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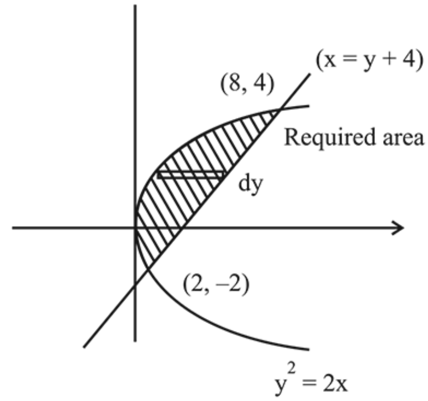
Q10

(4)



Given region, $A = \left\{(x,y) : \frac{y^2}{2} \leq x \leq y+4\right\}$

Hence, area $= \int_{-2}^{4} x\,dy = \int_{-2}^{4}\left(y+4-\frac{y^2}{2}\right)dy$

$= \left[\frac{y^2}{2}+4y-\frac{y^3}{6}\right]_{-2}^{4} = \left(8+16-\frac{64}{6}\right)-\left(2-8+\frac{8}{6}\right)$

$= \left(24-\frac{32}{3}\right)-\left(-6+\frac{4}{3}\right) = \frac{40}{3}+\frac{14}{3} = \frac{54}{3} = 18$

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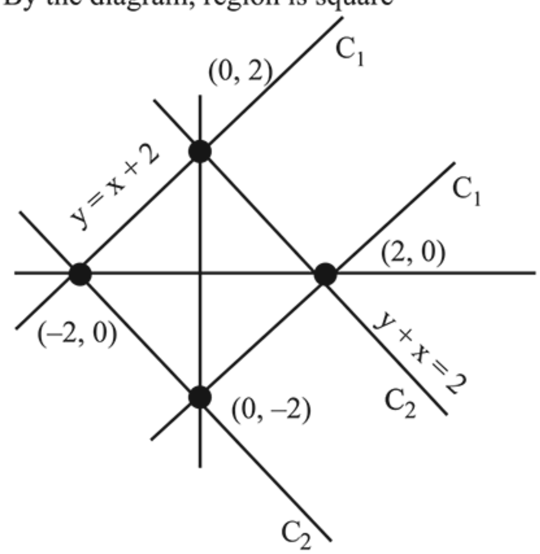
Q11

(1) Let, $C_1 : |y-x| \leq 2$

$C_2 : |y+x| \leq 2$

By the diagram, region is square



Now, length of side $= \sqrt{2^2+2^2} = 2\sqrt{2}$

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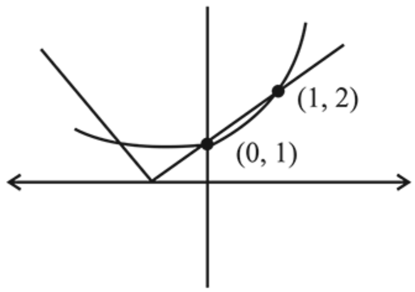
Q12

(4)



$$\text{Area} = \int_0^1 ((x+1) - 2^x)dx \qquad \left(\because \text{Area} = \int y dx\right)$$

$$= \left[\frac{x^2}{2} + x - \frac{2^x}{\ln 2}\right]_0^1 = \left(\frac{1}{2} + 1 - \frac{2}{\ln 2}\right) - \left(\frac{-1}{\ln 2}\right) = \frac{3}{2} - \frac{1}{\ln 2}$$

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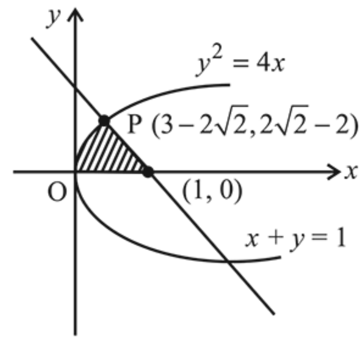
Q13

(2) Consider $y^2 = 4x$ and $x + y = 1$



Substituting $x = 1 - y$ in the equation of parabola,

$y^2 = 4(1 - y) \Rightarrow y^2 + 4y - 4 = 0$

$\Rightarrow \quad (y + 2)^2 = 8 \Rightarrow y + 2 = \pm 2\sqrt{2}$

Hence, required area

$$= \int_0^{3-2\sqrt{2}} 2\sqrt{x}\, dx + \frac{1}{2} \times (2\sqrt{2} - 2) \times (2\sqrt{2} - 2)$$

$$= \left[2 \times \frac{2}{3} x^{3/2}\right]_0^{3-2\sqrt{2}} + \frac{1}{2}(8 + 4 - 8\sqrt{2})$$

$$= \frac{4}{3} \times (3 - 2\sqrt{2})\sqrt{3 - 2\sqrt{2}} + 6 - 4\sqrt{2}$$

$$= \frac{4}{3}(3 - 2\sqrt{2})(\sqrt{2} - 1) + 6 - 4\sqrt{2}$$

$$[\because (\sqrt{2} - 1)^2 = 3 - 2\sqrt{2}]$$

$$= \frac{4}{3}(3\sqrt{2} - 3 - 4 + 2\sqrt{2}) + 6 - 4\sqrt{2}$$

$$= -\frac{10}{3} + \frac{8}{3}\sqrt{2} = a\sqrt{2} + b$$

$$\therefore \quad a = 8/3 \text{ and } b = -10/3 \Rightarrow a - b = \frac{10}{3} + \frac{8}{3} = 6$$

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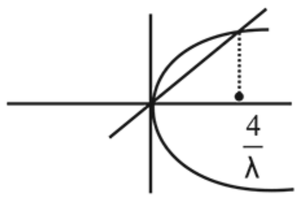
Q14

(3) Given parabola $y^2 = 4\lambda x$ and the line $y = \lambda x$



Putting $y = \lambda$ in $y^2 = 4\lambda x$, we get $x = 0, \dfrac{4}{\lambda}$

$\therefore$ required area $= \displaystyle\int_0^{\frac{4}{\lambda}} (2\sqrt{\lambda x} - \lambda x)dx$

$$= \frac{2\sqrt{\lambda}.x^{3/2}}{3/2} - \frac{\lambda x^2}{2}\Bigg|_0^{4/\lambda} = \frac{32}{3\lambda} - \frac{8}{\lambda}$$

$$= \frac{8}{3\lambda} = \frac{1}{9} \Rightarrow \lambda = 24$$

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